

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain with diagram in brief TTT curves.
- Q.24 Explain working principles and selection of furnaces for heat treatment operations.
- Q.25 Explain briefly Laser and electron beam hardening.

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5th Sem / Mechanical (Tool & Die Design)

Subject : Heat Treatment

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which is not common heat treatment process?
- a) Annealing b) Tempering
- c) Quenching d) Extrusion
- Q.2 External surface of part made of mild steel can be hardened by?
- a) Tempering b) Case hardening
- c) Normalizing d) Hardening
- Q.3 Nitriding is
- a) Brittle b) Malleable
- c) Ductile d) Tough

Q.4 Main purpose of Annealing?

- a) Improve machinability
- b) Increase hardness
- c) Improve magnetism
- d) Increase Toughness

Q.5 Which is not a type of carburizing?

- a) pack b) liquid
- c) gas d) Induction

Q.6 Heat treatment process used for softening hardened steel is

- a) Carburizing b) annealing
- c) normalizing d) tempering

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define Allotropy

Q.8 What is Soaking?

Q.9 What is SLA steel?

Q.10 Define electron beam hardening.

Q.11 Define flame hardening

Q.12 Write various quenching media of heat treatment

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain types of case hardening?

Q.14 Explain principles of heat treatment?

Q.15 Explain eutectoid points of curves?

Q.16 Explain in brief classification of steel?

Q.17 Explain micro structural changes of annealing process

Q.18 Compare Hardening & tempering process?

Q.19 Explain Tool steels with few examples?

Q.20 Explain Die steels with few applications?

Q.21 Explain gas carburizing

Q.22 Explain cryogenic Quenching.